

INDUSTRIAL INTERNSHIP REPORT

"Prediction of Agriculture Crop Production in India"

Prepared by

Jagruti Jiralikar

B.Tech Computer Engineering | MIT Academy of Engineering, Alandi, Pune

Internship Domain	Data Science & Machine Learning
Organisation	UpSkill Campus & UniConverge Technologies Pvt. Ltd.
Internship Period	17 June 2026 – July 2026 (6 Weeks)
Mode	Remote
GitHub Repository	github.com/jiralikarjagruti-svg/crop-production-prediction
Contact	jiralikarjagruti@gmail.com

Executive Summary

This report provides details of the Industrial Internship provided by UpSkill Campus.

TABLE OF CONTENTS

1. Preface	3
2. Introduction	4
2.1 About UniConverge Technologies Pvt. Ltd.....	4
2.2 About UpSkill Campus (USC)	4
2.3 Objectives.....	4
2.4 References	5
2.5 Glossary	5
3. Problem Statement.....	6
4. Existing and Proposed Solution.....	7
4.1 Existing Solutions and Their Limitations	7
4.2 Proposed Solution.....	7
4.3 Code Submission (GitHub Link)	7
4.4 Report Submission (GitHub Link)	8
5. Proposed Design / Model.....	9
5.1 High Level Pipeline Diagram.....	9
5.2 Low Level Design — Model Architecture	9
5.3 Interfaces — Streamlit Web Application	10
6. Performance Test	11
6.1 Test Plan / Test Cases.....	11
6.2 Test Procedure.....	11
6.3 Performance Outcome	11
7. My Learnings	13
8. Future Work Scope	14

1. Preface

This report presents the complete work carried out during a six-week Industrial Internship programme offered by UpSkill Campus in collaboration with UniConverge Technologies Pvt. Ltd. The internship was conducted remotely in the domain of Data Science and Machine Learning, commencing 17 June 2026.

The project undertaken was titled '**Prediction of Agriculture Crop Production in India**' — a regression-based machine learning project using real historical government data covering 29 Indian states, 124 crop types, and 6 seasons across the years 1997 to 2015 (246,091 records).

Over six weeks, the complete data science pipeline was implemented independently: dataset acquisition, data cleaning and quality validation, exploratory data analysis (EDA), feature engineering, training and comparing three machine learning models, building a deployed Streamlit web application, and maintaining a clean, well-documented GitHub repository. Each week's progress was submitted as a formal progress report on the UpSkill Campus platform.

This internship provided hands-on exposure to real-world data quality issues, model selection methodology, deployment constraints, and professional engineering practices that go well beyond typical academic coursework. I am grateful to the UpSkill Campus team and UniConverge Technologies for creating this structured learning opportunity.

To fellow students and juniors: theoretical knowledge and practical application are two different skills. Start building real projects early — you will learn more from one messy real-world dataset than from ten textbook examples. This internship proved that.

2. Introduction

2.1 About UniConverge Technologies Pvt. Ltd.

UniConverge Technologies Pvt. Ltd. (UCT), established in 2013, is a digital transformation company providing industrial solutions with a focus on sustainability and return on investment. UCT leverages cutting-edge technologies including IoT, Cyber Security, Cloud Computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, and front-end frameworks.

- **UCT Insight (IoT Platform)** — Designed for quick deployment of IoT applications. Built in Java (backend) and ReactJS (frontend), with support for MySQL and NoSQL databases. Supports device connectivity via MQTT, CoAP, HTTP, Modbus TCP, and OPC UA. Features include custom dashboards, analytics, alerting, and third-party integration (Power BI, SAP, ERP).
- **Smart Factory Platform** — A scalable solution for production and asset monitoring, OEE measurement, predictive maintenance, and digital twin capabilities for factory assets.
- **LoRaWAN-based Solutions** — UCT is an early adopter of LoRaWAN technology, providing solutions in AgriTech, Smart Cities, Industrial Monitoring, Smart Street Lighting, and Smart Metering.
- **Predictive Maintenance** — Industrial machine health monitoring and predictive maintenance solutions leveraging Embedded Systems, Industrial IoT, and Machine Learning to find the remaining useful lifetime of machines in production processes.

2.2 About UpSkill Campus (USC)

UpSkill Campus (USC), in association with UniConverge Technologies and The IoT Academy, facilitated the smooth execution of this internship programme. USC is a career development platform that delivers personalised executive coaching in an affordable, scalable, and measurable way. USC aims to upskill one million learners over the next five years and provides services including structured internships, industry expert interactions, project-based learning, and career growth support.

2.3 Objectives

- Gain practical, hands-on experience working on an industry-relevant data science problem.
- Solve a real-world prediction problem using machine learning techniques on real data.
- Train and compare multiple models using a scientific, evidence-based approach to model selection.
- Deploy the best-performing model as a user-facing interactive web application.
- Develop professional engineering habits: version control (Git/GitHub), structured documentation, and modular code organisation.
- Build understanding of the complete ML workflow — from raw data to a deployed solution.

2.4 References

- Kaggle — Crop Production in India Dataset (abhinand05):
<https://www.kaggle.com/datasets/abhinand05/crop-production-in-india>
- scikit-learn Documentation — RandomForestRegressor, LabelEncoder, train_test_split:
<https://scikit-learn.org>
- Streamlit Documentation: <https://docs.streamlit.io>
- pandas Documentation: <https://pandas.pydata.org>
- joblib Documentation: <https://joblib.readthedocs.io>
- GitHub Repository: <https://github.com/jiralikarjagruti-svg/crop-production-prediction>

2.5 Glossary

Term / Acronym	Definition
ML	Machine Learning
EDA	Exploratory Data Analysis — visualising and understanding raw data
R ² Score	Coefficient of Determination — measures model accuracy (0 to 1, higher is better)
MAE	Mean Absolute Error — average prediction error in tonnes (lower is better)
Random Forest	Ensemble ML model using multiple decision trees to improve accuracy
Label Encoding	Converting categorical text values into numeric format for ML models
Streamlit	Python library for building interactive data science web applications
joblib	Python library for serialising and deserialising Python objects (e.g. ML models)
GitHub	Cloud-based platform for hosting and version-controlling code repositories
Git	Distributed version control system for tracking code changes
Kharif	Indian monsoon cropping season (June–October)
Rabi	Indian winter cropping season (October–March)

3. Problem Statement

Agriculture is the backbone of India's economy, contributing approximately 18% to GDP and employing over 50% of the workforce. Accurately forecasting crop production is critical for food security planning, supply-chain management, price stabilisation, and agricultural policy-making.

However, crop production is influenced by a complex combination of factors — geographic region, crop variety, cultivated area, seasonal weather patterns, and year-on-year variations — making accurate prediction a non-trivial problem.

The assigned problem was to:

- Build a machine learning regression model capable of predicting the total production (in tonnes) of a specified crop in a given Indian state.
- Train the model on real government historical data covering 29 states, 124 crop types, and 6 seasons from 1997 to 2015.
- Accept the following inputs: State Name, Crop Year, Season, Crop Type, and Cultivated Area (in hectares).
- Output a predicted Production value (in tonnes) with an estimated confidence range.
- Deliver the solution as an interactive web application accessible to non-technical users.

An additional real-world data quality challenge was encountered during the project: the dataset contained unit inconsistencies (Coconut production recorded in nuts/count rather than tonnes) and an incomplete year (2015 contained only 561 records vs. ~13,000 average for other years). Identifying and handling these issues was part of the problem scope.

4. Existing and Proposed Solution

4.1 Existing Solutions and Their Limitations

Prior approaches to agricultural production prediction have largely relied on:

- **Simple Linear Regression:** Assumes a linear relationship between cultivated area and production. In practice, this relationship is highly non-linear across different crops and states. As confirmed in this project, Linear Regression achieved only an R^2 score of 0.074 (7.4%) on this dataset.
- **Statistical time-series models:** These focus only on year-over-year trends and do not account for the categorical variables (crop type, season, state) that are the strongest predictors of production.
- **Lack of interactive tools:** Most existing solutions are standalone scripts or notebooks with no user-facing interface, making them inaccessible to agricultural officers, policymakers, or farmers.
- **Data quality blind spots:** Published analyses rarely address unit mismatches (e.g., Coconut in nuts vs. other crops in tonnes) or incomplete yearly records, which can severely distort analysis results if unaddressed.

4.2 Proposed Solution

The proposed solution is a complete, end-to-end ML pipeline using a **Random Forest Regressor** — an ensemble model capable of capturing complex non-linear relationships and handling mixed data types (categorical + numerical). The solution addresses all limitations identified above:

- Rigorous data cleaning and quality validation to handle missing values, whitespace inconsistencies, unit mismatches, and incomplete yearly records.
- Label Encoding to systematically convert categorical features (State, Season, Crop) into numeric inputs suitable for ML model training.
- Systematic training and comparison of three regression models (Linear Regression, Decision Tree, Random Forest) using identical train-test splits, evaluated by R^2 score and Mean Absolute Error.
- A deployed, interactive Streamlit web application with a professional three-tab layout (Predict / Historical Trends / Model Comparison) that non-technical users can operate without any programming knowledge.
- All code, data, and model files hosted on a public GitHub repository with version control and a .gitignore configuration.

4.3 Code Submission (GitHub Link)

<https://github.com/jiralikarjagruiti-svg/crop-production-prediction>

4.4 Report Submission (GitHub Link)

<https://github.com/jiralikarjagruti-svg/crop-production-prediction>

5. Proposed Design / Model

5.1 High Level Pipeline Diagram

The project followed a standard, industry-aligned machine learning pipeline:

Stage	Activity	Output / Deliverable
1. Data Acquisition	Downloaded dataset from Kaggle	crop_production.csv (246,091 rows, 7 columns)
2. Data Cleaning	Removed missing Production rows; Fixed whitespace in Crop names	242,361 clean records
3. EDA	Analysed top crops, states, seasons, year trends, correlations	5 visualisation charts; 2 data quality findings
4. Feature Engineering	Label Encoding on 3 categorical cols; 80/20 train-test split	X_train (192,322 rows) X_test (48,081 rows)
5. Model Training	Trained & compared 3 regression models	Best: Random Forest $R^2=0.958$, MAE=8,638 t
6. Model Saving	Saved model + encoders with joblib compression (compress=3)	crop_model.pkl (17 MB) (reduced from 1.5 GB)
7. Web Application	Built Streamlit app: 3 tabs, custom dark theme, download btn	Live interactive app localhost:8501
8. GitHub Push	Committed all files with meaningful commit messages	Public repository (all code + model hosted)

5.2 Low Level Design — Model Architecture

The final model is a **Random Forest Regressor** with the following configuration and data flow:

Component	Details
Algorithm	Random Forest Regressor (ensemble of decision trees)
n_estimators	50 trees (reduced from 100 to optimise file size)
max_depth	15 levels per tree (controls model complexity / overfitting)
random_state	42 (ensures reproducibility across runs)
Input Features (5)	State_encoded, Crop_Year, Season_encoded, Crop_encoded, Area
Target Variable	Production (in tonnes)
Training set	192,322 rows (80% of 240,403 usable records)
Test set	48,081 rows (20% of usable records)

Serialisation

```
joblib.dump(model, 'crop_model.pkl', compress=3) → 17 MB
```

5.3 Interfaces — Streamlit Web Application

The trained model was wrapped in a **Streamlit web application** with three tabs:

- **Tab 1 — Predict:** Two-column input layout with dropdown selectors for State, Season, and Crop, plus numeric inputs for Year and Area. On clicking 'Predict Production', the app returns the predicted production value, an estimated confidence range (derived from the 50 individual tree predictions within the Random Forest), an input summary table, and a CSV download button for the prediction report. A feature importance bar chart is permanently displayed below.
- **Tab 2 — Historical Trends:** Interactive line chart showing year-wise production history for any selected state-crop combination (1997–2015). Displays three summary metrics: average production, highest recorded production, and total record count.
- **Tab 3 — Model Comparison:** Side-by-side bar charts comparing R^2 Score and MAE across Linear Regression, Decision Tree, and Random Forest. A detailed results table and an explanatory insight note explain why Random Forest was selected.

The application uses a **custom dark theme** with agriculture-themed green accents (primary colour #4CAF50) configured via Streamlit's *config.toml*, and includes developer credits and a copyright footer.

6. Performance Test

6.1 Test Plan / Test Cases

All three candidate models were evaluated on identical conditions to ensure a fair, scientific comparison:

- **Same dataset split:** 80% training (192,322 rows) and 20% testing (48,081 rows), using `random_state=42` for reproducibility.
- **Same input features:** `State_encoded`, `Crop_Year`, `Season_encoded`, `Crop_encoded`, `Area`.
- **Same evaluation metrics:** R^2 Score (primary) and Mean Absolute Error / MAE (secondary).
- **Three models tested:** Linear Regression (baseline), Decision Tree Regressor (intermediate), and Random Forest Regressor (ensemble).

6.2 Test Procedure

- Each model was instantiated with default or specified parameters, then fitted on `X_train`, `y_train`.
- Predictions were generated on `X_test` using `model.predict(X_test)`.
- R^2 Score was calculated using `sklearn.metrics.r2_score(y_test, y_pred)`.
- MAE was calculated using `sklearn.metrics.mean_absolute_error(y_test, y_pred)`.
- Feature importance was extracted from the Random Forest using `model.feature_importances_`.
- Results were stored in a dictionary and exported as `model_comparison.json` for display in the web app.

6.3 Performance Outcome

Model	R^2 Score	MAE (tonnes)	Assessment
Linear Regression	0.074 (7.4%)	64,592	Poor — high bias on non-linear data
Decision Tree	0.799 (79.9%)	10,717	Good — captures non-linearity
Random Forest ✓	0.958 (95.8%)	8,638	Excellent — selected as final model

Key Findings:

- The Random Forest Regressor achieved $R^2 = 0.958$, correctly explaining 95.8% of variance in crop production — significantly above the industry benchmark of 0.85+.
- Feature importance analysis: **Crop type (60.8%)** is the dominant predictor, followed by Area (26.0%), State (9.3%), Year (2.9%), and Season (1.0%). This confirms that what you grow matters more than where or when.

- Linear Regression's failure ($R^2=0.074$) validates the dataset's non-linear nature and justifies the choice of an ensemble model — a finding directly supported by the bias-variance theory studied in the Week-4 ML modules.
- Model optimisation ($n_estimators=50$, $max_depth=15$, $compress=3$) reduced file size from **1.5 GB to 17 MB** while maintaining the same R^2 of 0.958 — a 98.9% size reduction with zero loss in accuracy.
- Two critical data quality issues were identified and resolved: (a) Coconut unit mismatch (recorded in nuts, not tonnes) — excluded from cross-crop comparisons; (b) Incomplete 2015 records (561 rows vs ~13,000 average) — excluded from trend analysis.

7. My Learnings

This six-week internship provided a concentrated, hands-on learning experience across the full data science workflow. Below are the most significant learnings from each phase of the project:

- **Real-world data is always messy.** The dataset had missing values in the target column, extra whitespace in category names, unit mismatches across crops, and an incomplete final year. Identifying and handling these issues before modelling is as critical as the model itself — and is a skill rarely taught in academic settings.
- **Model selection must be evidence-based, not intuitive.** Training three models and comparing them on identical splits revealed that Linear Regression's simplicity is a liability on non-linear data ($R^2=0.074$), while Random Forest's ensemble approach captured complex patterns effectively ($R^2=0.958$). This reinforced that the right model is discovered through experimentation, not assumed.
- **Feature importance transforms predictions into insights.** Discovering that Crop type (60.8%) and Area (26%) together account for 87% of the prediction power is not just a modelling result — it is an actionable agricultural insight: crop choice drives production far more than location or timing.
- **Engineering constraints are real, and problem-solving them is a skill.** A 1.5 GB model file was a deployment blocker. Resolving it through parameter reduction and joblib compression (achieving 17 MB at identical accuracy) was a practical engineering problem not covered in any course — and one that every practitioner eventually faces.
- **Deployment bridges ML and impact.** A model in a notebook is a proof of concept. A model in a Streamlit web app with tabs, confidence ranges, download buttons, and a professional UI is a tool that a real agricultural officer could use. Building for the user — not just for accuracy — is a mindset shift this project helped develop.
- **Version control is a professional hygiene skill.** Using Git throughout the project — with meaningful commit messages, a .gitignore, and a clean folder structure — directly mirrors industry practice and is a habit that employers actively look for.
- **Theory and practice reinforce each other.** Studying ensemble methods and the bias-variance tradeoff in the Week-4 ML modules made the project results more interpretable. Conversely, having seen the results first made the theory intuitive. This bidirectional learning is the core value of project-based internships.

8. Future Work Scope

The following enhancements were identified during the project but could not be implemented within the six-week timeframe. They represent concrete directions for future iterations:

- **Hyperparameter Tuning:** Applying GridSearchCV or RandomizedSearchCV to the Random Forest model could further optimise `n_estimators`, `max_depth`, `min_samples_split`, and `max_features` parameters, potentially pushing the R^2 score above 0.96.
- **Incorporating Weather and Soil Data:** The current model relies only on administrative and agricultural variables. Merging with historical rainfall, temperature, and soil quality data would add the most important real-world predictors of crop yield that are absent from the current dataset.
- **District-Level Predictions:** The `District_Name` column is available in the dataset but was not included as a feature in this model. Adding district-level granularity would enable significantly more localised and actionable predictions.
- **Crop Recommendation System:** Extending the project from a prediction tool ('given this crop, how much will it produce?') to a recommendation system ('given this state and area, which crop should I grow for maximum yield?') would dramatically increase the tool's real-world agricultural value.
- **Online Deployment:** Deploying the Streamlit application to Streamlit Community Cloud or AWS would make the tool publicly accessible via a permanent URL without requiring local installation, enabling real-world stakeholder use.
- **Time-Series Forecasting:** Replacing the regression approach with a time-series model (e.g., LSTM, Prophet) to forecast future years' production trends for specific crops and states — enabling forward-looking agricultural planning rather than retrospective analysis.
- **Cross-Validation:** Implementing k-fold cross-validation ($k=5$ or 10) would provide a more statistically robust estimate of model generalisation performance than a single train-test split.

GitHub Repository (Code + Report):
<https://github.com/jiralikarjagruiti-svg/crop-production-prediction>

© 2026 Jagruti Jiralikar | Industrial Internship Report | UpSkill Campus × UniConverge Technologies Pvt. Ltd.